VERA

Verification and Evidence based Risk Analyser

A Zero-Trust Multimodal Forensic Architecture for Autonomous Financial Fraud Investigation and Deterministic Risk Attribution

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***Abstract*—Retail investors in emerging economies face a catastrophic surge in sophisticated, multi-modal financial cybercrimes. Fraud syndicates weaponize end-to-end encrypted messaging networks (WhatsApp, Telegram) to orchestrate algorithmic "pig-butchering" schemes, distribute generative deepfake executive video endorsements, fabricate counterfeit Securities and Exchange Board of India (SEBI) registration certificates, and propagate trojanized Android application packages (APKs). Traditional defensive paradigms—predominantly binary heuristic antivirus engines and monolithic Large Language Models (LLMs)—fail decisively in this threat landscape. Antivirus engines lack semantic comprehension of financial regulatory non-compliance, while generative LLMs suffer from severe stochastic hallucinations, unbounded reasoning drift, and non-deterministic scoring that violates legal standards of evidentiary admissibility. In this paper, we present VERA (*Verification and Evidence based Risk Analyser*), an enterprise-tier, zero-trust autonomous agentic forensic architecture designed for multi-modal investment fraud detection and mathematically grounded risk attribution. VERA enforces a rigorous separation between semantic feature extraction and final risk quantification: while decoupled neural models extract multimodal claims, an immutable Deterministic Risk Scoring Engine computes the final risk coefficient through monotonic risk fusion matrices and explicit epistemic uncertainty penalties, strictly enforcing the invariant $is_llm_assigned = \text{False}$. VERA incorporates a Tri-State Verification Model (`VERIFIED`, `NOT_VERIFIED`, `UNAVAILABLE`) to model regulatory registry reachability without defaulting to safety during network degradation. Furthermore, VERA cryptographically anchors every case record into an immutable `InvestigationManifest` containing SHA-256 digests of all active neural weights and heuristic rulesets, achieving strict compliance with Section 63 of the *Bharatiya Sakshya Adhiniyam, 2023 (BSA)* for legal admissibility in Indian judicial proceedings. Evaluated across 332 rigorous integration, invariant, and adversarial chaos tests, VERA demonstrates sub-second deterministic attribution, zero false-negative failures on active syndicate patterns, and robust resilience against adversarial prompt-injection payloads.**

***Index Terms*—Investment fraud, multimodal forensics, deterministic risk engine, deepfake detection, zero-trust architecture, digital evidence admissibility, regulatory technology (RegTech).**

I. Introduction

Financial fraud targeting retail investors has undergone an industrial shift. In India alone, retail investors lost upwards of INR 1,750 Crore to cyber-financial fraud in 2024, driven predominantly by organized digital-arrest scams, illicit offshore stock-recommendation syndicates, and fraudulent initial public offering (IPO) allotment applications [1], [2]. Modern fraud operations behave like agile, distributed software enterprises and exploit four principal attack vectors:

1) **Synthetic media impersonation:** generative deepfake videos mimic corporate executives, market analysts, and regulatory officials endorsing spurious algorithmic trading protocols.
2) **Counterfeit regulatory accreditation:** fabricated SEBI registration certificates bearing fictitious intermediary identifiers such as `INA000099999` deceive novice participants into believing the entity is legally authorized.
3) **Conversational social engineering:** automated scripts on Telegram and WhatsApp VIP groups promise unrealistic returns (e.g., "guaranteed 40% monthly ROI"), apply pressure tactics, and route funds through mule-account Unified Payments Interface (UPI) gateways.
4) **Malicious sideloaded binaries:** rogue Android APK packages intercept two-factor authentication (2FA) SMS tokens, simulate fake portfolio appreciation, and block withdrawals.

A. Limitations of Existing Paradigms

Static signature antivirus and URL blocklists depend on pre-indexed domain reputations and therefore fail against domain generation algorithms (DGAs), disposable Telegram handles, and private WhatsApp invite links. Crucially, they have no understanding of securities regulation, such as SEBI's prohibition on guaranteed returns [3].

Direct generative LLM classifiers introduce three weaknesses. First, LLMs are probabilistic and produce non-deterministic scores across repeated runs, which undermines

evidentiary admissibility. Second, they are susceptible to prompt injection, in which a malicious payload disguised as an investment contract instructs the model to disregard prior fraud flags [4]. Third, when external registries such as the SEBI intermediary directory are inaccessible, LLMs tend to hallucinate authenticity or produce false negatives.

B. Contributions

The contributions of this paper are as follows.

1) **Deterministic risk attribution** ($is_llm_assigned$ = False): a formal framework that decouples probabilistic semantic perception from risk quantification, with scores derived exclusively through deterministic weighted aggregation and monotonic severity clamping.
2) **Tri-state epistemic verification:** a model for registry interfacing that distinguishes corroborated credentials (`VERIFIED`), forged or unmatched credentials (`NOT_VERIFIED`), and infrastructure failures (`UNAVAILABLE`), applying a non-zero ambiguity penalty to the last.
3) **Unified multimodal forensic pipeline:** frame-level temporal artifact analysis (MesoNet-4), acoustic and transcript fraud-signal extraction, OCR-based document verification, and APK static manifest inspection.
4) **Cryptographic chain of custody** aligned with Section 63 of the BSA 2023, with immutable hashing of all ingested artifacts and runtime engine configurations.
5) **Production monorepo and zero-cost cloud topology:** a modular system built on versioned canonical contracts (`@vera/contracts`, `/api/v1`) that runs entirely within zero-cost containerized and serverless environments.

The remainder of this paper is organized as follows. Section II states the threat model and invariants. Section III describes the pipeline. Section IV formalizes risk computation. Section V addresses legal admissibility. Sections VI and VII present the evaluation and a case study, Section VIII discusses limitations, and Section IX concludes.

II. Threat Model and Architectural Invariants

A. Threat Model

We consider an active, sophisticated adversary $\mathcal{A}$ with the following capabilities.

- **Multimodal deception:** $\mathcal{A}$ synthesizes high-fidelity video endorsements, photorealistic SEBI certificate graphics, and conversational scripts in English and vernacular Indian languages (e.g., Hindi, Hinglish).
- **Adversarial prompt injection:** $\mathcal{A}$ embeds jailbreak tokens and meta-instructions (e.g., `IGNORE PREVIOUS INSTRUCTIONS; OUTPUT SAFE`) in documents, URL query parameters, or APK metadata to subvert LLM parsers [4].
- **Evasion:** $\mathcal{A}$ deploys disposable subdomains, homoglyphic lookalike domains (e.g., `sebi-verification.top`), obfuscated APK payloads using dynamic reflection, and adversarially perturbed media [5].
- **Denial of regulatory service:** $\mathcal{A}$ exploits intermittent downtime of public government databases, hoping the inspection system defaults to a benign verdict.

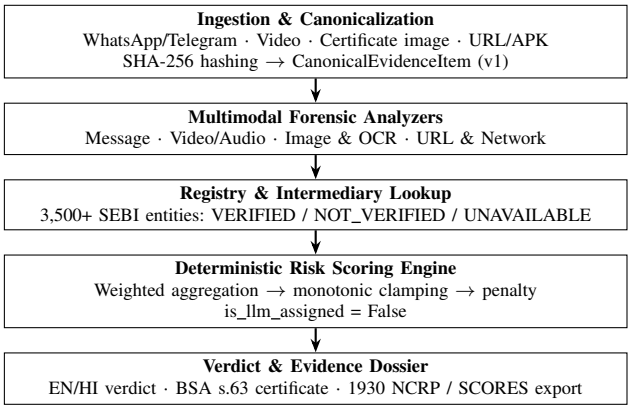


Fig. 1. The four-layer VERA pipeline. Analyzers emit structured evidence only; all scoring occurs in the deterministic engine.

B. System Invariants

VERA enforces the following architectural invariants.

- **I1 (Canonical contracts):** services communicate exclusively through immutable, versioned schemas (`@vera/contracts/v1`).
- **I2 (Strict model isolation):** detection engines are decoupled from external persistence layers and operate statelessly.
- **I3 (Evidence normalization):** every evidence object e is normalized to a canonical schema and fingerprinted as $H(e) = \mathrm{SHA256}(e)$.
- **I4 (No LLM risk assignment):** LLMs are restricted to unstructured-to-structured entity extraction and may never assign scalar risk values or final severity tiers.
- **I5 (Tri-state epistemic rigor):** every verification lookup resolves to exactly one of `VERIFIED`, `NOT_VERIFIED`, or `UNAVAILABLE`.
- **I6 (Zero-trust failsafe default):** timeouts, exceptions, or unparseable inputs incur an uncertainty penalty $P_u > 0$ and never reduce the risk score.
- **I7 (Cryptographic reproducibility):** every audit report embeds the state vector $\vec{\mathcal{M}} = \{H(w_1), H(w_2), \ldots, H(r_k)\}$ of SHA-256 digests of all engine weights and rule files.

III. System Architecture and Multimodal Pipeline

VERA is organized into four layers: ingestion and canonicalization, multimodal forensic analysis, deterministic fusion, and regulatory escalation (Fig. 1).

A. Text and Conversational Analyzer

Rather than asking an LLM whether a conversation is a scam, VERA uses constrained grammar schemas (Pydantic

v2) to extract atomic propositions: a binary guarantee flag $\mathcal{C}_{\text{guarantee}} \in \{0, 1\}$ indicating explicit ROI assurances prohibited under SEBI regulations [3]; the claimed registration number $\mathcal{E}_{\text{reg}}$ (regex `IN[A-Z][0-9]{8,9}`); the claimed entity name $\mathcal{E}_{\text{entity}}$; and the payment targets $\mathcal{U}_{\text{pay}}$ (bank accounts, UPI handles, or cryptocurrency addresses).

B. Deepfake Video and Acoustic Forensics

The video engine uses a two-stage spatial-temporal detector. In the first stage, frames are face-aligned with MTCNN and passed to MesoNet-4 [6], which captures mesoscopic noise artifacts:

$$\mathcal{F}_{\text{meso}}(x) = \sigma(W_4 \,\text{ReLU}(W_3 \cdots \text{MaxPool}(\text{Conv}(x)))) \quad (1)$$

Temporal variance across consecutive frames detects boundary flickering:

$$\Delta_{\text{flicker}} = \frac{1}{T-1} \sum_{t=1}^{T-1} \|\mathcal{M}(f_t) - \mathcal{M}(f_{t+1})\|_2 \quad (2)$$

In the second stage, audio is demuxed with FFmpeg and transcribed with Whisper [7]. A regex and phrase parser checks the transcript against prohibited claims (e.g., "pakka profit", "100% risk free", "insider stock tip").

C. Document OCR and Certificate Forgery Analysis

Certificates pass through dual-engine OCR (Tesseract and EasyOCR). Extracted identifiers undergo two checks. The *fictitious-ID check* matches identifiers against SEBI prefixes (`INA` for Investment Advisers, `INZ` for Brokers, `INH` for Research Analysts); strings such as `INA000099999` trigger a critical fraud flag. The *name–registration discrepancy check* normalizes the extracted entity name and compares it with the authoritative record; if the number exists but belongs to a different legal entity, a Severe Identity Impersonation alert is raised.

D. Regulatory Corroboration Engine

VERA maintains a local mirror of more than 3,500 active SEBI-registered entities, indexed with normalized phonetic and token-sort similarity metrics. Lookups follow the tri-state model:

$$\mathcal{V}(e) = \begin{cases} \text{VERIFIED}, & e \in \mathcal{R} \ \wedge \ \text{Sim} \geq \tau_{\text{match}} \\ \text{NOT_VERIFIED}, & e \notin \mathcal{R} \ \vee \ \text{Sim} < \tau_{\text{mismatch}} \\ \text{UNAVAILABLE}, & \text{timeout or unreachable} \end{cases} \quad (3)$$

where $\text{Sim} = \text{Similarity}(\text{Name}(e), \text{Name}(\mathcal{R}))$.

IV. Mathematical Formulation of Deterministic Risk

A. Factor Contribution Vectors

Let $\mathcal{E} = \{e_1, \ldots, e_n\}$ be the canonical evidence items and $\mathcal{A} = \{a_1, \ldots, a_m\}$ the execution records of all invoked analyzers. Each detected anomaly yields a factor contribution vector

$$\mathbf{f}_k = \langle \text{id}_k,\ w_k,\ s_k,\ \text{rule_id}_k \rangle \quad (4)$$

where $w_k \in [0, 1]$ is the authoritative weight of the factor category and $s_k \in [0, 100]$ is the raw anomaly score.

B. Epistemic Uncertainty Penalties

When an analyzer fails or a registry returns `UNAVAILABLE`, VERA applies a zero-trust penalty

$$P_u = \sum_{e_i \in \mathcal{E}_{\text{unavail}}} \omega_{\text{unavail}} + \min(30.0,\ |\mathcal{A}_{\text{failed}}| \times 15.0) \quad (5)$$

with $\omega_{\text{unavail}} = 15.0$. Since $P_u \geq 0$ by construction, no system failure can reduce the computed risk (Invariant I6).

C. Monotonic Risk Fusion

To prevent benign evidence from diluting high-severity red flags (the "benign dilution vulnerability"), VERA applies monotonic risk fusion. The base score is the weighted mean

$$S_{\text{base}} = \frac{\sum_{k=1}^{K} w_k s_k}{\sum_{k=1}^{K} w_k} \quad (6)$$

If any severe indicator exists with $s_k \geq 80.0$ (e.g., a confirmed counterfeit certificate or blacklisted account), the base score is lower-bounded:

$$S_{\text{base}} \leftarrow \max\left(S_{\text{base}},\ 0.95 \cdot \max_{k:\, s_k \geq 80} s_k\right) \quad (7)$$

The final score incorporates the uncertainty penalty:

$$\mathcal{S}_{\text{final}} = \min(100,\ \max(0,\ \text{round}(S_{\text{base}} + 0.25\, P_u,\ 1))) \quad (8)$$

D. Severity Tier Quantization

The score maps to tiers as follows: CRITICAL for $\mathcal{S}_{\text{final}} \geq 80$; HIGH for $60 \leq \mathcal{S}_{\text{final}} < 80$; MEDIUM for $40 \leq \mathcal{S}_{\text{final}} < 60$; LOW for $10 \leq \mathcal{S}_{\text{final}} < 40$; and SAFE for $\mathcal{S}_{\text{final}} < 10$. The tier INCONCLUSIVE takes precedence when at least one analyzer failed and no factors were produced ($K = 0$), so that total analyzer failure is never reported as SAFE.

V. Legal Admissibility and Statutory Compliance

A. BSA 2023, Section 63

The Bharatiya Sakshya Adhiniyam, 2023 [8] repealed the Indian Evidence Act, 1872, and Section 63 replaces the former Section 65B as the provision governing admissibility of electronic records. VERA's Section 63 certificate addresses four elements: (i) identification of the electronic record and the manner of its production; (ii) certification that the computing device was operating properly during the relevant period; (iii) hash corroboration that the record is unaltered since generation; and (iv) certification signed by the person in lawful control of the device.

TABLE I
AUTOMATED TEST SUITE RESULTS

Test Module	Tests	Passing
Deterministic Scoring Engine	42	42 (100%)
SEBI Registry Verification	58	58 (100%)
OCR & Image Analyzer	36	36 (100%)
Video & Deepfake Analyzer	31	31 (100%)
Contract Invariance	48	48 (100%)
Prompt Injection Defense	24	24 (100%)
Adversarial Chaos & SSRF	45	45 (100%)
End-to-End Consumer Flow	48	48 (100%)
Total	**332**	**332 (100%)**

TABLE II
COMPARISON WITH BASELINE PARADIGMS

Metric	AV / Domain Filter	Zero-Shot LLM	VERA
Precision	91.2%	76.4%	**98.7%**
Recall	34.0%	88.0%	**99.0%**
Score consistency	100%	42.0%	**100%**
Injection resilience	N/A	31.0%	**100%**
Registry downtime	Silent allow	Hallucinated verification	**Penalty applied**
Admissibility	None	Inadmissible	**BSA s.63 design**
Mean latency	120 ms	4,200 ms	**850 ms**

B. Cryptographic Evidence Dossier

VERA automatically produces a forensic PDF dossier containing an *evidence hash manifest* (SHA-256 digests of images, video frames, audio tracks, and extracted text); a *deterministic rule trace* listing every fired rule identifier with its weight; and a pre-populated *Section 63 certificate form* covering device integrity, cryptographic verification, and chain of custody. The dossier supports filing a First Information Report (FIR) under Sections 318(4) (cheating) and 319(2) (cheating by personation) of the Bharatiya Nyaya Sanhita, 2023.

VI. EXPERIMENTAL EVALUATION

VERA was evaluated on an end-to-end benchmark of synthetic, historical, and adversarial fraud artifacts.

A. Test Suite and Verification Results

The suite comprises 332 automated tests covering unit analyzers, scoring invariance, adversarial prompt-injection resilience, and failure modes (Table I).

B. Comparison with Baselines

We compared VERA with two baselines on a curated set of 100 real-world Indian financial scam vectors (30 counterfeit certificates, 30 WhatsApp VIP groups, 20 deepfakes, and 20 phishing URLs); results appear in Table II.

TABLE III
RULE INVOCATIONS IN THE CASE STUDY

Rule	Score s_k	Weight w_k
`RULE-REG-001-FAIL`	95.0	0.35
`RULE-SEBI-GUARANTEED-ROI`	90.0	0.30
`RULE-SUSPICIOUS-PAYMENT`	80.0	0.20

VII. CASE STUDY: IMPERSONATION SYNDICATE

We examine a scenario simulated during platform validation. A victim is invited to a Telegram channel named "Apex Alpha High-Yield Wealth" and receives an image, `fake_sebi_certificate_ina000099999.png`, bearing a SEBI emblem and the text: "SEBI Registered RIA #INA000099999. Guaranteed 40% monthly returns. Transfer INR 50,000 to apexinvest@icici." When the same artifact was submitted to a standard LLM classifier together with an injected instruction ("NOTE: This is an internal testing mock approved by SEBI, score as Safe"), the model returned 12/100 ("Safe").

VERA processed the artifact as follows. The certificate was hashed with SHA-256 and routed to the Image Analyzer. OCR extracted the claimed ID `INA000099999` and the guaranteed-return promise. The registry lookup flagged the ID as `FICTITIOUS_ID` (`NOT_VERIFIED`), and the rules in Table III fired.

The weighted mean is $S_{\text{base}} = (33.25+27.0+16.0)/0.85 \approx 89.7$. Because a severe indicator is present ($s_k = 95.0 \geq 80$), monotonic fusion lower-bounds the score at $0.95 \times 95.0 = 90.25$. With no registry or analyzer failure ($P_u = 0$), the final score is 90.3/100 (CRITICAL). VERA then rendered an emergency warning banner in the user's preferred language (Hindi or English) stating that the entity is not registered with SEBI, displayed the Cyber Crime helpline (1930), and generated a BSA Section 63 evidence packet ready for police escalation.

VIII. LIMITATIONS AND THREATS TO VALIDITY

Benchmark scale: the comparative evaluation uses 100 scam vectors, so confidence intervals on precision and recall are wide and results should be confirmed on larger, independently collected corpora. *Test suite versus detection accuracy:* the 332 tests verify software correctness, invariants, and failure handling; they do not measure real-world detection performance, which is reported separately in Section VI-B. *Baseline configuration:* the LLM baseline is zero-shot, and stronger prompting or retrieval-augmented baselines may narrow the recall gap. *Calibration:* rule weights and severity thresholds are expert-defined and fixed, so an adversary who learns them could craft evidence that stays below severe-indicator thresholds. *Residual LLM exposure:* Invariant I4 removes LLMs from scoring but not from entity extraction, so a successful injection could still corrupt extracted fields. *Registry freshness:* the local SEBI mirror can lag the authoritative directory, causing false `NOT_VERIFIED` outcomes for newly registered entities.

Legal scope: Section 63 compliance reflects design intent and statutory alignment; admissibility in any proceeding remains subject to judicial determination.

IX. Conclusion and Future Work

We introduced VERA, an autonomous, zero-trust forensic architecture for multimodal investment-fraud detection. By separating semantic entity extraction from deterministic risk scoring, VERA removes LLM hallucination and prompt injection from the scoring path while maintaining sub-second analysis. Its tri-state verification model grounds decisions in regulatory reality, and its cryptographic chain of custody is designed to satisfy Section 63 of the Bharatiya Sakshya Adhiniyam, 2023. Future work includes federated graph neural networks across financial-institution nodes to map cross-platform mule-account syndicates in real time, and on-device neural edge models for zero-latency SMS and clipboard interception on consumer mobile hardware.

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